

Core Definitions

Artificial Intelligence (AI)

Definition: Artificial Intelligence encompasses computer systems capable of performing tasks that normally require human intelligence, including perception, reasoning, learning, problem-solving, and language understanding.

Core Components:

- Data analysis and pattern recognition
- Reasoning and problem-solving
- Natural language processing
- Learning from experience

Real-world Applications: Chatbots, fraud detection, medical diagnosis, recommendation engines, autonomous vehicles.

Common Interview Questions & Answers:

Q: What is Artificial Intelligence?

A: AI is the field of computer science focused on creating systems or agents that can perform tasks typically requiring human intelligence, such as learning, reasoning, and perception.

Q: What are the types of AI?

A:

- Artificial Narrow Intelligence (ANI): Specialized for a particular task
- Artificial General Intelligence (AGI): Can perform any intellectual task as a human
- Artificial Superintelligence (ASI): Surpasses human intelligence

Similarity with ML/DL: All are aiming to create intelligent systems; ML and DL are subfields of AI.

Machine Learning (ML)

Definition: A subset of AI, Machine Learning enables computers to learn from data and improve from experience without being explicitly programmed for every scenario.

Machine Learning Process:

- Define objective (problem statement)
- Gather and prepare data (cleansing, normalization)
- Select and train algorithms
- Evaluate model performance (accuracy, recall, F1-score)
- Deploy and monitor predictions

Advantages: Adapts to new data, automates decision-making, scalable solutions.

Types of Machine Learning:

- Supervised Learning (with labels)
- Unsupervised Learning (without labels)
- Reinforcement Learning (reward-based learning)

Common Interview Questions & Answers:

Q: What is Machine Learning?

A: ML is a branch of AI that focuses on building systems that can learn from data and make decisions or predictions.

Q: How is ML different from AI?

A: ML is a subset of AI that specifically focuses on algorithms that learn from data, whereas AI is a broader concept encompassing any technique that enables computers to mimic human intelligence.

Differences (ML vs. AI):

Aspect	AI (Broad)	ML (Subset)
Scope	Encompasses reasoning, planning, etc.	Focuses on learning from data and predictions
Goal	Mimic human intelligence	Allow systems to learn, adapt, and predict
Example	Chess computer, voice assistant	Spam detection, credit scoring

Deep Learning (DL)

Definition:

Deep Learning is a subset of ML leveraging multi-layered neural networks capable of extracting complex features from raw data, excelling in domains like image and speech recognition, NLP, and more

Key Components:

- Neural Networks (layers of nodes/neurons)
- Activation Functions (e.g., ReLU, Sigmoid)
- Training with Backpropagation
- Convolutional Neural Networks (CNNs): Best for images
- Recurrent Neural Networks (RNNs): Best for sequences/text
- GANs, LSTMs for specialized tasks

Advantages: Automated feature extraction, state-of-the-art results for unstructured data, scalable to huge datasets.

Disadvantages:

Requires massive labeled datasets, computationally intensive, less interpretable ("black box") compared to traditional ML.

Common Interview Questions & Answers:

Q: What is Deep Learning?

A: Deep Learning is a branch of ML that uses neural networks with multiple layers to automatically learn representations of data and solve complex tasks.

Q: How does Deep Learning differ from Machine Learning?

A:

- DL uses neural networks with many layers, can automatically extract features, and is suited for large/unstructured data.
- ML can include less complex models, usually needs manual feature engineering, and works with smaller datasets.

Comparison Table (ML vs. DL):

Aspect	Machine Learning	Deep Learning
Data Requirement	Moderately large	Very large
Feature Eng.	Manual	Automatic (learns from data)
Interpretability	Easier to interpret	Often a black box
Hardware Need	Moderate	High (GPUs often needed)
Application	Tabular data, predictions	Images, speech, NLP, and complex data

Supervised Learning

Definition: Models trained using datasets with known input-output pairs (labels). The model learns to map inputs to correct outputs

Common Algorithms: Linear regression, logistic regression, decision trees, SVMs, neural networks.

Applications: Spam classification, disease diagnosis, sentiment analysis.

Common Questions & Answers:

- Q: What is supervised learning?
A: It's a learning paradigm in which the model is trained on labeled data to predict outputs for unseen data.
- Q: Give examples.
A: Email spam detection, loan approval, medical image diagnosis.

Unsupervised Learning

Definition: Learning from unlabeled data, discovering hidden structure or patterns¹.

Common Algorithms: K-Means clustering, PCA (Principal Component Analysis), Apriori (association rules).

Applications: Customer segmentation, anomaly detection, data compression.

Common Questions & Answers:

- Q: What is unsupervised learning?
A: Learning from unlabeled data to find patterns or groupings without predefined outputs.

- Q: Examples?
A: Clustering customers by behavior, reducing dimensions in datasets.

Differences (Supervised vs. Unsupervised):

Aspect	Supervised	Unsupervised
Labels	Required	Not required
Tasks	Classification, Regression	Clustering, Association
Evaluation	Accuracy, RMSE, F1	Silhouette score, inertia

Reinforcement Learning (RL)

Definition: An agent learns to act in an environment to maximize cumulative reward, using feedback in the form of rewards/penalties

Process:

- Agent interacts with environment
- Takes action → gets reward/penalty
- Learns best sequence of actions (policy)

Applications: Robotics, game playing (e.g., AlphaGo), recommendation systems.

Common Questions & Answers:

Q: What is reinforcement learning?

A: A learning paradigm where an agent learns by performing actions and receiving feedback in the form of rewards or penalties, refining its strategy (policy) over time.

Online vs. Offline Learning

Definition:

- Online Learning: Model is updated incrementally as each new data point arrives; ideal for streaming or continually updating data
- Offline Learning (Batch): Model is trained on the entire dataset at once; not updated until retraining with a new dataset

Differences Table:

Aspect	Online Learning	Offline (Batch)
Data Flow	Streams/new instances	All data at once
Adaptation	Continuous	Retraining needed
Use Cases	Stock trading, IoT	Medical image analysis

Similarity: Both seek to improve model performance with data; main difference is data handling and adaptation frequency.

Batch vs. Stochastic Learning (SGD)

Definition:

- Batch Learning: Updates model parameters after passing through the entire dataset at once. Suitable for smaller or fixed datasets.
- Stochastic Learning (SGD): Updates model parameters after each data sample (or mini-batch)—enables faster, scalable, and continual learning, especially for large datasets.

Comparison Table:

Aspect	Batch Learning	Stochastic Learning (SGD)
Data Processing	Entire dataset	One or few points at a time
Convergence	Stable, slower	Faster, noisier
Memory Use	High	Low
Scalability	Limited	High

Interview Q & A:

Q: When would you use batch learning vs. stochastic learning?

A: Use batch learning for small to medium static datasets with sufficient computational resources. Use SGD/mini-batch for very large or streaming datasets, where you need faster updates and lower memory use.

Key Concepts and Terminology

- Features: Individual measurable properties used as input for modeling.
- Labels: (Supervised learning) The target outputs/answers in the training set.
- Preprocessing: Cleaning, normalizing, transforming data prior to modeling.
- Overfitting: When a model learns noise/specifically training data and fails to generalize.
- Bias & Ethics: Addressing fairness, transparency, and unintended consequences in AI/ML systems